DO NOW — RECALL

Lesson 067 · Divide and Conquer: Splitting the Search

Name: _____ Class: _____ Date: _____

1. Define decomposition.

2. Which input assumption does binary search require?

3. How does Math.Min(a,b) behave?

4. Binary search [2,4,6,8,10,12,14] for 12: show low/high and midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Breaking a complex problem into smaller, manageable parts. The parts may depend on one another.

2. Model answer

Sorted elements.

3. Model answer

Returns the smaller argument.

4. Model answer

low0 high6 mid3=8, then low4 high6 mid5=12; found index5.

By the end, you can…

• Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

• Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

• Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Vocabulary — know these terms

divide and conquer — Solve smaller subproblems and combine.

precondition — Requirement for correctness.

binary search — Halving search on sorted data.

merge — Combine sorted partial results.

NEW LEARNING

Read, annotate and explain

Define and predict

Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

Explain

Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

Evaluate

Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

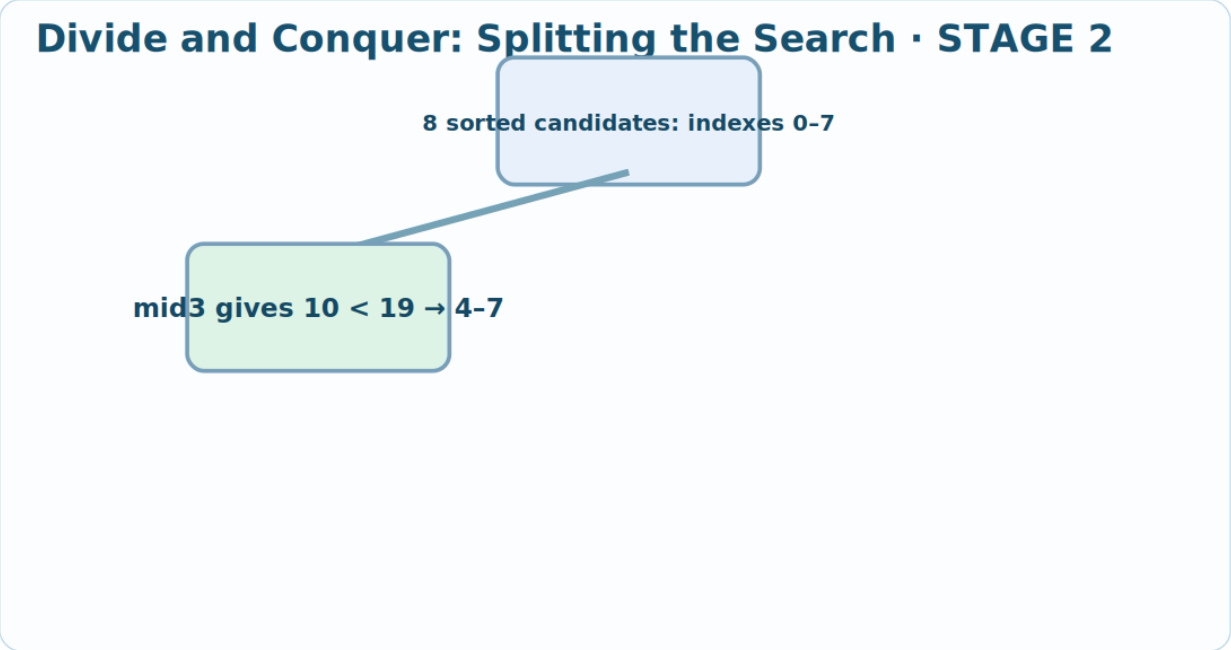


Divide and Conquer: Splitting the Search · STAGE 1; 8 sorted candidates: indexes 0–7

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

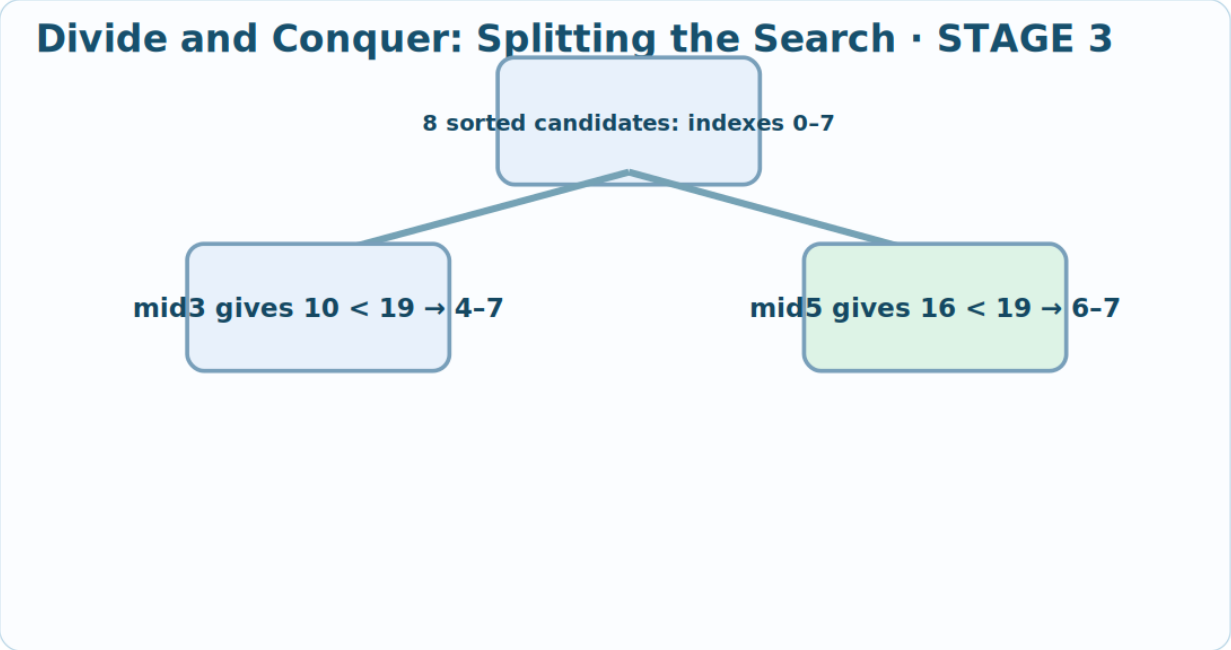


Divide and Conquer: Splitting the Search · STAGE 2; 8 sorted candidates: indexes 0–7; mid3 gives 10 < 19 → 4–7

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

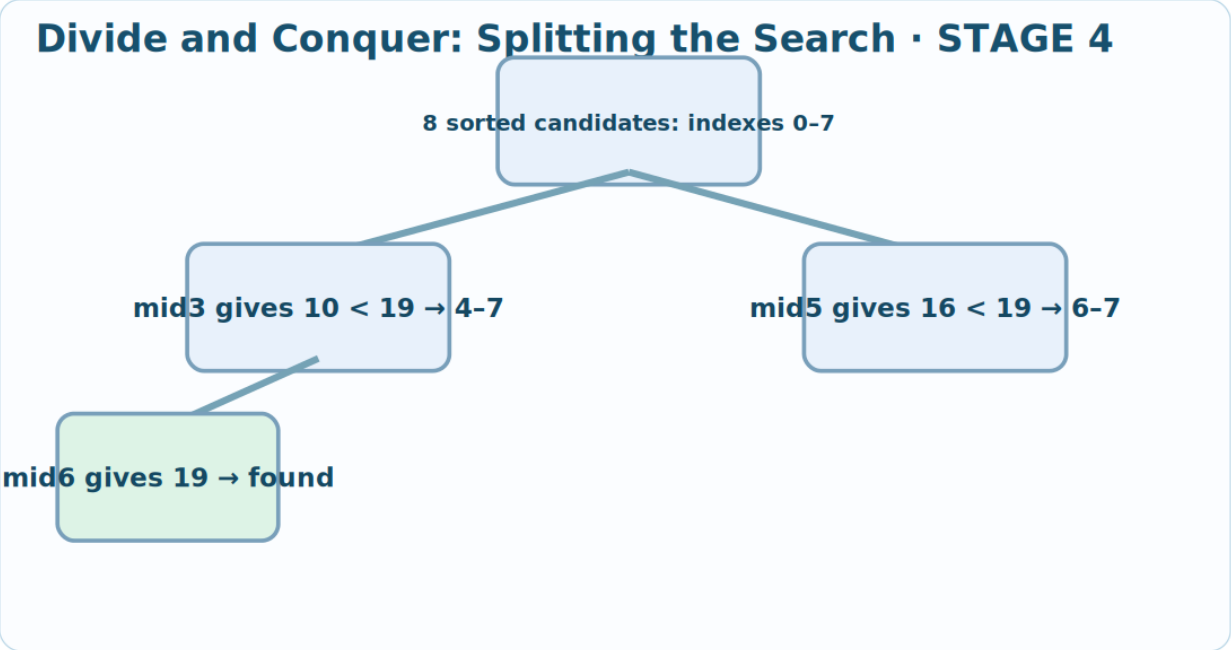


Divide and Conquer: Splitting the Search · STAGE 3; 8 sorted candidates: indexes 0–7; mid3 gives 10 < 19 → 4–7; mid5 gives 16 < 19 → 6–7

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Divide and Conquer: Splitting the Search · STAGE 4; 8 sorted candidates: indexes 0–7; mid3 gives 10 < 19 → 4–7; mid5 gives 16 < 19 → 6–7; mid6 gives 19 → found

I DO — WORKED EXAMPLE

Follow the worked method

Search sorted [2,5,8,11,14,17,20,23] for 20.

Follow the reasoning

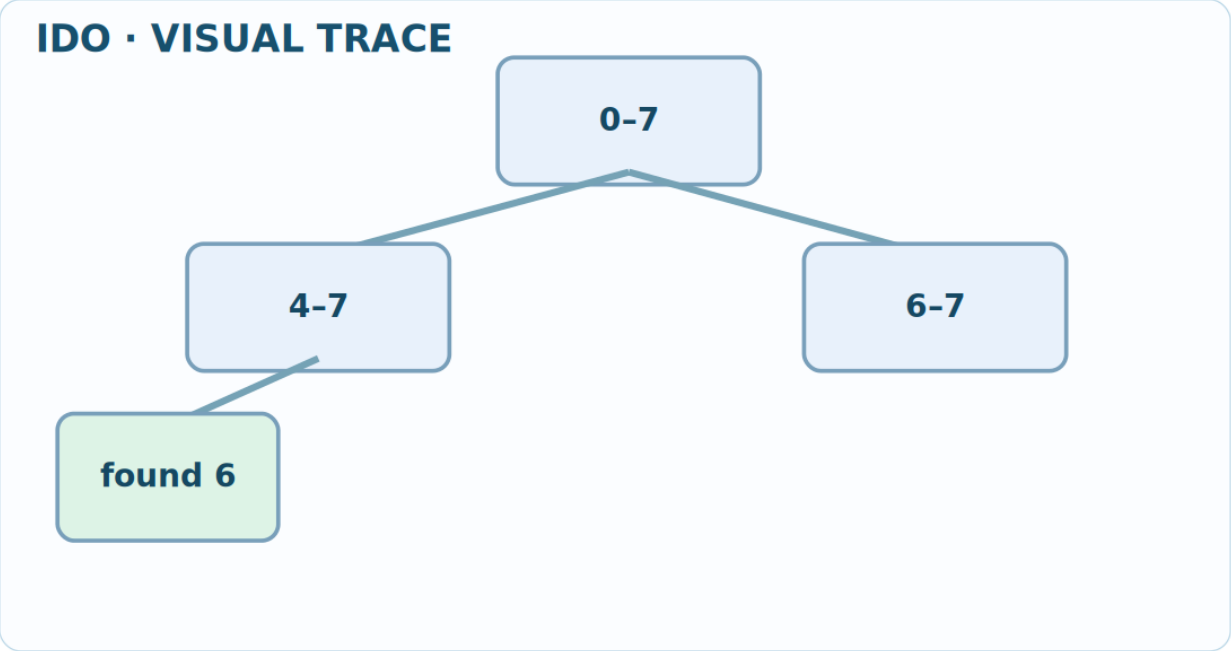
Each iteration narrows the candidate interval, and result is index 6.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Search sorted [2,5,8,11,14,17,20,23] for 20.

Each iteration narrows the candidate interval, and result is index 6.



IDO · VISUAL TRACE; 0–7; 4–7; 6–7; found 6

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Search sorted [3,6,9,12,15,18,21,24] for 6.
Source code / input:
mid3=12 → left indexes0–2; mid1=6 → found at index1.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For [1,4,7,10,13,16,19,22] trace middle indexes when target19. [6]

Question 2

Explain divide, solve and combine for merge sort on [7,2,9,1]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare repeated halving with linear scan for sorted list of 1024 items, in terms of worst-case comparisons and preconditions. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Binary search will correctly find any target in any unsorted list.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

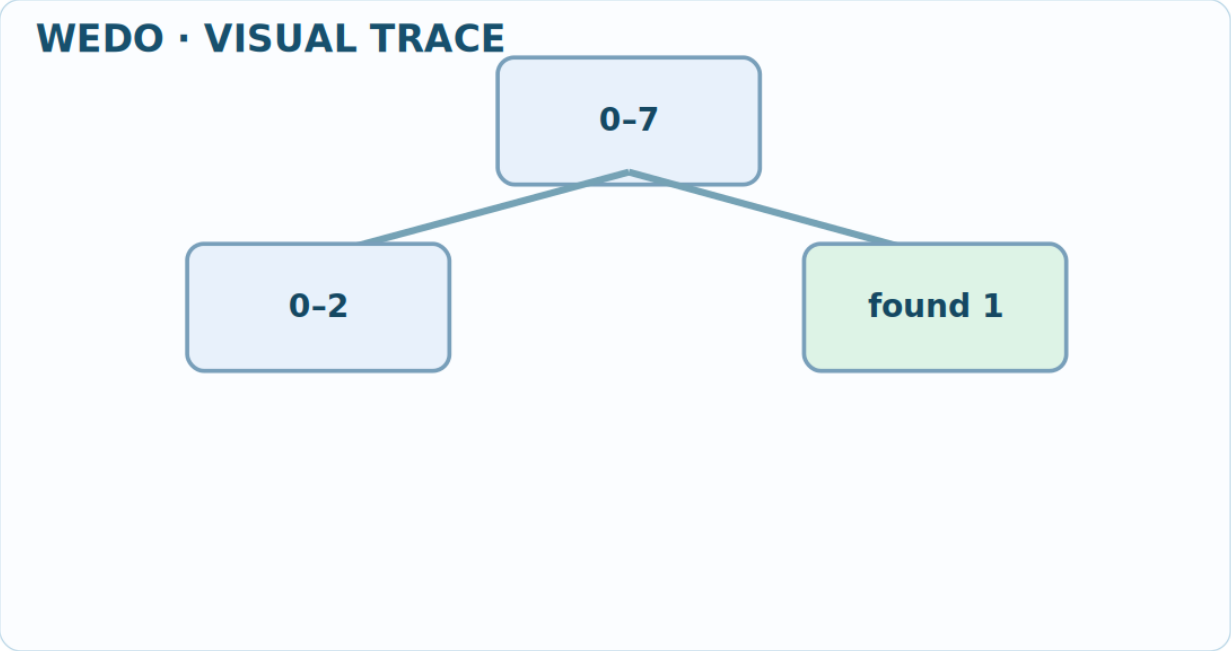
Follow the reasoning
The second comparison finds the target after removing upper half.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Search sorted [3,6,9,12,15,18,21,24] for 6.

The second comparison finds the target after removing upper half.



WEDO · VISUAL TRACE; 0–7; 0–2; found 1